Mini Project (CSE7102)
Review-0 (10^{th} August 2026)

PRJ-314 AI-Enabled Water Well Predictor

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Abstract

Access to reliable groundwater is a critical component of sustainable infrastructure, directly aligned with UN Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure). Traditional well-site assessment relies heavily on manual hydrogeological surveys, which are time-consuming, resource-intensive, and often inaccessible in under-resourced regions. This project proposes an AI-enabled water well predictor that leverages historical groundwater monitoring data, along with associated rainfall and temperature records, to forecast groundwater levels across multiple stations in India.

The system uses a multistation spatiotemporal dataset spanning 1994–2025, incorporating groundwater levels, rainfall, and temperature as core features. After preprocessing (handling missing values, feature engineering for seasonal and lagged climate effects), the project applies machine learning regression models — beginning with a Random Forest/Gradient Boosting baseline — to predict groundwater levels at a given location and time. These predictions are further translated into an interpretable well-risk classification (e.g., Safe, Declining, Critical), enabling practical decision-making for well-site viability and water resource planning.

The final deliverable includes a lightweight web-based interface where a user can input a region and time period to receive a predicted groundwater level along with a corresponding risk category. By combining time-series climate data with modern machine learning techniques, this project aims to deliver a scalable, low-cost decision-support tool for groundwater management, without dependency on physical sensors or IoT infrastructure — improving automation and real-world usability while keeping the solution accessible and reproducible.